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Leveraging Large Language Model for Enhanced Text-to-SQL Parsing

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ABSTRACT Text-to-SQL conversion, the process of generating SQL queries from natural language input, has gained significant attention due to its potential to simplify database interaction. Although benchmarks in this task have driven advancements in the field, the challenges posed by complex join logic and the rich diversity of natural language expressions remain significant obstacles. These complexities underscore the ongoing difficulty of accurately bridging the gap between natural language and structured query representations, particularly in cross-domain and real-world scenarios. Recent research, including intermediate representations, relation-aware transformers, and large language models such as T5 and LLaMA, has improved performance by addressing the semantic gap between natural language and SQL. In this work, we propose SLENet, a novel approach that uses state-of-the-art large language models (LLMs) to enhance semantic understanding and SQL generation. Our method integrates three core innovations: (1) the use of advanced LLMs for context-aware representations, (2) syntax-constrained SQL decoder to ensure grammatical correctness, and (3) search-based prompt optimization utilizing external knowledge sources like WikiSQL. These innovations collectively address schema comprehension and SQL generation complexities. Evaluations on the Spider benchmark demonstrate that SLENet significantly outperforms existing methods, achieving higher exact matching accuracy and effectively handling complex SQL components. Our contributions highlight the importance of combining LLMs with syntax constraints and external data for advancing cross-domain semantic parsing.

INDEX TERMS Semantic Parsing, SQL Generation, Deep Learning, LLM, Deep Learning.

I. INTRODUCTION

Text-to-SQL conversion, the task of synthesizing SQL queries from natural language inputs to query relational databases, has gained considerable attention due to its potential to make database interaction more accessible and intuitive [1]– [5]. Public benchmarks such as ATIS, GeoQuery, WikiSQL, and Spider have demonstrated that existing SQL synthesis approaches can achieve high exact matching accuracy (e.g., exceeding 85%). However, the increasing complexity of real-world datasets, characterized by cross-domain schemas, intricate SQL structures, and unseen databases, continues to pose significant challenges. These complexities underscore the need for advancing techniques to effectively bridge the gap between natural language and structured query representations. Recent advancements in neural models have made substantial progress in addressing these challenges. Techniques such as IRNet [5] and RAT-SQL [7] utilize intermediate representations and relation-aware transformers to resolve discrepancies between natural language syntax and SQL structures, while models like SmBoP [8] and PICARD [4] employ advanced decoding strategies to enhance SQL generation and ensure syntactic correctness. Pre-trained language models such as T5 [9] and Codex [10] have further set new performance benchmarks by leveraging transfer learning and few-shot learning capabilities, demonstrating their ability to generalize across diverse datasets. Additionally, advancements in hierarchical schema representations (e.g., HSRNet) and intermediate parsing techniques have improved schema comprehension and simplified SQL generation, particularly in cross-domain scenarios.

Despite these technological strides, critical challenges remain. Current methods still struggle to handle the diversity of natural language expressions, the complexity of multi-table queries, and the demand to generalize to unseen database

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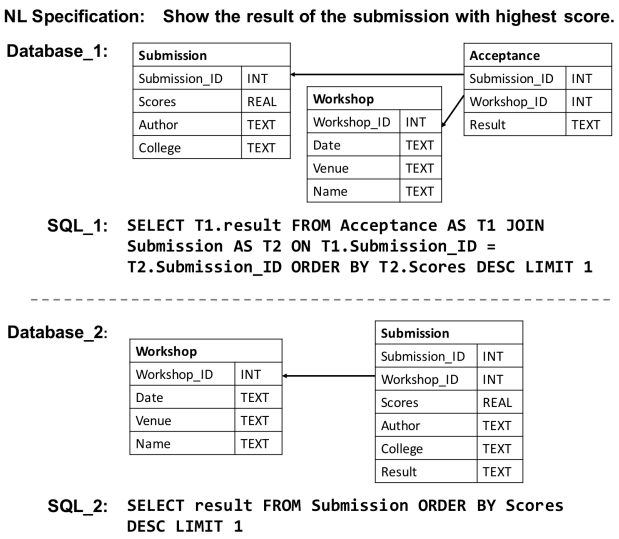

FIGURE 1. Example input NL specification x, corresponding SQL queries y_{sql}, and the related database schema s. The first example demonstrates a complex query requiring cross-table joins, while the second focuses on sorting and selecting attributes. These examples illustrate the challenges of schema complexity and SQL query generation.

schemas. These limitations highlight the need for continued innovation, as existing approaches often fall short in fully bridging the semantic gap between natural language and SQL in real-world applications. The Spider benchmark [6], which emphasizes realistic and complex scenarios, serves as a critical evaluation metric, highlighting these ongoing challenges and the need for robust solutions.

To address these challenges, we propose SLENet, a novel approach that leverages state-of-the-art large language models (LLMs) to advance Text-to-SQL conversion. SLENet introduces three core innovations: (1) context-aware representations through advanced LLMs, (2) a syntax-constrained SQL decoder to ensure grammatical correctness, and (3) search-based prompt optimization leveraging external knowledge sources. These contributions build upon recent advancements in cross-domain semantic parsing and neural architectures, such as execution-guided decoding for real-time query validation and retrieval-augmented generation for integrating external data. Evaluations on the Spider benchmark demonstrate that SLENet significantly outperforms existing methods, achieving higher exact matching accuracy and effectively handling complex SQL components. This work marks a substantial step toward overcoming the persistent challenges in Text-to-SQL conversion and advancing the field.

II. APPROACH

In this section, we provide an in-depth explanation of our proposed approach, SLENet, which is organized into three interconnected modules: Search-Based Prompt, LLM Enhanced Embedding, and Syntax-Constrained Generation. These components work in tandem to address the challenges of schema comprehension and SQL generation.

A. METHOD OVERVIEW

Given a natural language (NL) input $x = x_1 \cdots x_{|x|}$ and a corresponding database schema S, our objective is to generate a syntactically correct and semantically meaningful SQL query $y_{\text{sql}} = y_{\text{sql}_1} \cdots y_{\text{sql}_{|y_{\text{sql}}|}}$. To achieve this, we leverage large language models (LLMs) to extract enriched information embeddings, complemented by a search-based prompt mechanism to enhance the quality of input representations. Inspired by prior work on grammar-based decoding [11], [12], we design a tree-structured syntax-constrained decoder to guide the query generation process. The framework is illustrated in Fig. II.

B. LLM ENHANCED EMBEDDING

To capture the relationship between the NL input x and the database schema S, we employ state-of-the-art LLMs to enhance semantic representations. The NL input and schema are enriched with a search-based prompt P derived from external data sources, creating a unified representation $I = \text{concat}(x, S, P)$. The LLM processes I to generate three key outputs: (1) an enhanced embedding for the NL input, E_{NL}; (2) a schema embedding, E_{S}; and (3) an initial SQL query y_{init}, which may contain errors due to the absence of syntactic constraints. The enriched representations, along with the preliminary SQL query, serve as input for the syntax-constrained decoder. This modular design ensures that semantic and structural information is effectively encoded while maintaining scalability for diverse Text-to-SQL tasks.

LLM Benchmark Models We adopt LLaMA 6B [13] as our benchmark model, which balances computational efficiency and strong performance across NLP tasks. Given an input query $X = x_1 \cdots x_{|X|}$, schema $S = s_1 \cdots s_{|S|}$, and search-based prompt $P = p_1 \cdots p_{|P|}$, the LLaMA model generates E_{NL}, E_{S}, and y_{init}. Its open-access nature and scalability make it an ideal choice for fine-tuning on domain-specific SQL generation tasks.

C. SYNTAX-CONSTRAINED DECODER

The decoder synthesizes SQL queries by adhering to predefined syntax rules, leveraging a grammar-based approach to ensure syntactic and semantic correctness. The generation process is formalized as:

$$p(y|x, s) = \prod_{i=1}^{T} p(a_i|x, s, a_{<i}),$$

where a_i is an action taken at time step i, $a_{<i}$ is the sequence of actions before i, and T is the total number of time steps in the action sequence.The decoder generates an SQL query through three types of actions: *APPLYRULE*, which applies production rules based on the abstract syntax tree (AST) of SQL grammar; *SELECTCOLUMN*, which selects a column c from the schema S using the input representation x and schema embedding E_S; and *SELECTTABLE*, which selects a table t from the schema S in a similar manner. These

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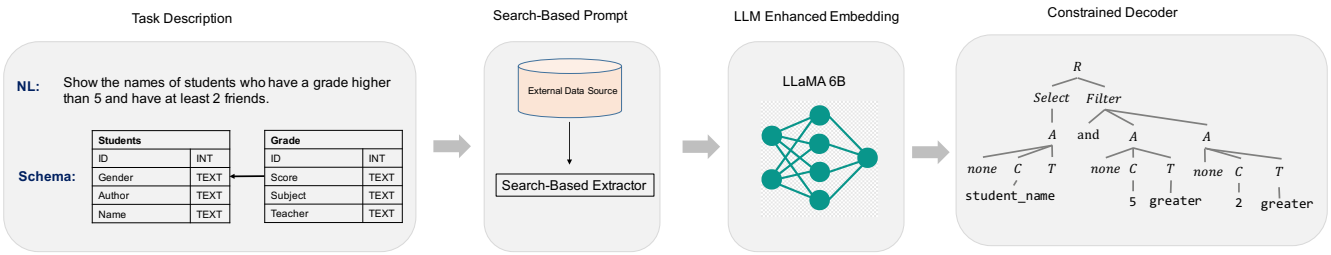

FIGURE 2. The figure illustrates the pipeline of our Text-to-SQL conversion approach, which integrates search-based prompt enrichment to incorporate external knowledge, LLM-enhanced embeddings to capture semantic and structural information from natural language inputs and database schemas, and a constrained decoder to generate syntactically correct SQL queries step by step.

actions enable the step-by-step construction of queries while ensuring syntactic and semantic correctness.

Each action is constrained by the SQL syntax rules and schema information, ensuring that the generated query is both valid and executable. The use of a grammar-based decoder guarantees that every step of the decoding process adheres to the structured nature of SQL. For further details on the APPLYRULE action, we refer readers to Yin and Neubig [11]. By incorporating these constrained actions, the decoder effectively generates accurate and executable SQL queries, while reducing the risk of syntax errors in the output.

III. MODEL

In this section, we present SLENet, a neural approach designed to generate SQL-specific grammar y_g from a natural language (NL) question and a database schema. SLENet addresses challenges in schema comprehension and SQL generation through two key components: the **LLM Enhanced Encoder** and **Constrained Decoder**. In the following subsections, we provide a detailed explanation of each component, including their roles, architectures, and the innovations they introduce to enhance SQL query generation.

A. LLM ENHANCED ENCODER

The proposed SLENet model integrates a natural language (NL) question, database schema, and a search-based prompt to enhance SQL generation. Leveraging the capabilities of LLaMA, the model extracts enriched embeddings for the NL input (E_x), schema (E_s), and contextually augmented columns (E_c). These embeddings are processed through a hierarchical structure to capture semantic and structural dependencies. The search-based prompt provides additional contextual information to guide SQL generation. The output embeddings are then fed into a constrained decoding module, ensuring the generation of SQL queries that adhere to grammar and schema constraints. This two-stage approach enhances both the semantic understanding and syntactic correctness of the SQL queries.

Searched-Based Prompt To enhance the input query x, we introduce external NL2SQL datasets and compute the relevance between the input query x and the external dataset queries Q. Specifically, the relevance is calculated as follows:

$$\text{Relevance}(x, q_i) = \text{Sim}(x, q_i)$$

where Q_i represents the i-th query in the external dataset Q, and $\text{Sim}(\cdot, \cdot)$ denotes a similarity function, such as cosine similarity or a neural embedding-based similarity. We then select the top N most relevant queries as prompt candidates. The external datasets used for this process can include standard NL2SQL datasets, such as WikiSQL [14], to provide diverse and contextually relevant prompt options for downstream tasks. This search-based prompt mechanism ensures that the model is guided by relevant, contextually similar examples, thereby improving the accuracy and relevance of thegenerated SQL queries.

Enhanced Encoder The encoder in SLENet takes as input the NL question, schema, and search-based prompt, as illustrated in Fig. III. To fully leverage the instruction-tuning capabilities of LLMs, we combine the NL question and the prompt into a single input sequence and specify the required task explicitly. The encoder is tasked with generating three outputs simultaneously: (1) token-level embeddings for the NL and schema, and (2) an initial SQL query that may contain errors due to the unconstrained generation process.

Formally, let the NL question be$X = \{x_1, x_2, \ldots, x_{|X|}\}$, the schema $S = \{s_1, s_2, \ldots, s_{|S|}\}$, and the prompt $P = \{p_1, p_2, \ldots, p_{|P|}\}$. The combined input to the encoder is represented as:

$$I = \text{concat}(P, X, S).$$

Using this input I, the LLM-based encoder generates the following outputs:

1. Token-Level NL and Schema Embeddings: The embeddings for each token in the NL and schema are represented as:

$$E_{\text{NL}} = \{e_{q_1}, e_{q_2}, \ldots, e_{q_{|Q|}}\}, \quad E_{\text{Schema}} = \{e_{s_1}, e_{s_2}, \ldots, e_{s_{|S|}}\}.$$

2. Initial SQL Query: The encoder also generates an initial SQL query Y_{init} directly:

$$Y_{\text{init}} = \text{LLM}(I),$$

where $y_{\text{init}} = \{y_{i1}, y_{i2}, \ldots, y_{i|y_{\text{sql}}|}\}$ represents the sequence of tokens in the generated SQL query. This query may

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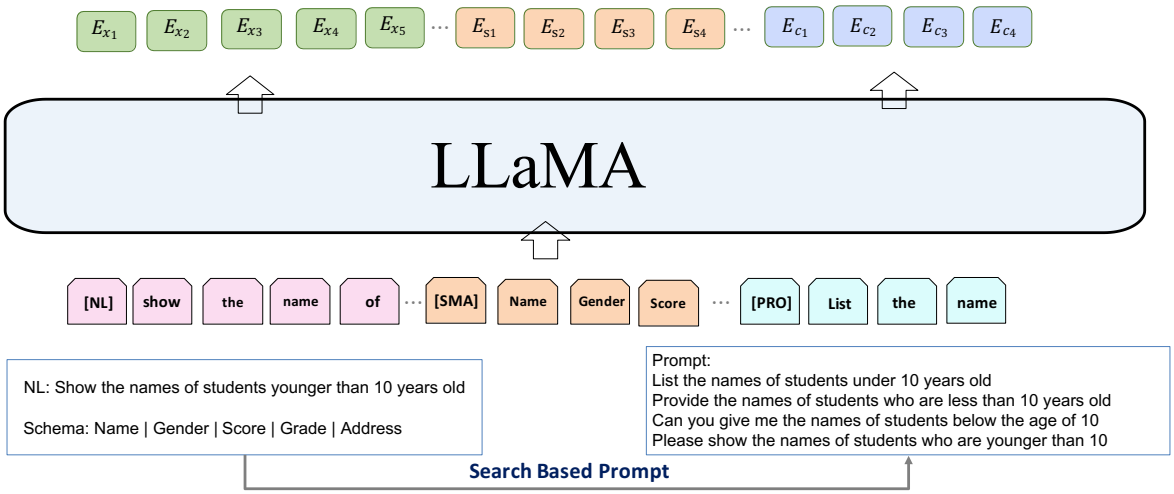

FIGURE 3. The figure illustrates the integration of a search-based prompt with the LLaMA model. The input consists of a natural language query, database schema, and contextually relevant prompts. LLaMA processes these inputs to generate enriched embeddings, capturing semantic and structural relationships for downstream SQL generation.

contain errors due to the lack of constraints in the initial generation process. The encoder concatenates the three outputs together as the input to the decoder.

$$E = \text{concat}(E_{\text{NL}}, E_{\text{Schema}}, Y_{\text{init}}).$$

B. CONSTRAINED DECODER

To generate SQL queries while adhering to SQL grammar rules, we adopt a syntax-constrained decoding strategy similar to IRNet [5]. Each step in the SQL generation process is restricted by the predefined SQL syntax, ensuring that the output remains syntactically valid. Unlike traditional approaches that use LSTMs [15], our method employs a transformer-based [16] model as the decoder, leveraging its powerful attention mechanism to capture long-range dependencies and semantic relationships effectively. The decoder processes the token embeddings produced by the LLM encoder and generates the SQL query token by token. The decoder consists of 4 layers, and each layer follows the standard transformer architecture, including positional embeddings, multi-head self-attention, and feed-forward networks (FFN). Additionally, SQL grammar constraints are applied through masking, ensuring that only valid tokens are considered at each decoding step.

Input and Positional Encoding. The input to the decoder is the concat sequence of token embeddings generated by the LLM encoder:

$$E = \{e_1, e_2, \ldots, e_{|E|}\},$$

where e_i represents the embedding of the i-th token. To capture positional information, we add positional embeddings to the input embeddings:

$$Z_0 = E + P,$$

where P is the positional encoding matrix.

Multi-Head Attention. At each decoder layer, the multi-head self-attention mechanism computes the attention-weighted representation of the input embeddings. For a given input Z_l at layer l, the self-attention output is computed as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V,$$

where $Q = Z_l W_Q$, $K = Z_l W_K$, and $V = Z_l W_V$ are the query, key, and value matrices derived from Z_l. W_Q, W_K, W_V are learned weight matrices, and d_k is the dimension of the key.The multi-head attention output is given by:

$$\text{MHA}(Z_l) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W_O,$$

where $\text{head}_i = \text{Attention}(Q_i, K_i, V_i)$ and W_O is the learned output weight matrix.

Feed-Forward Network (FFN). The output of the self-attention module is passed through a feed-forward network, which consists of two linear layers with a ReLU activation in between:

$$\text{FFN}(x) = \text{ReLU}(xW_1 + b_1)W_2 + b_2,$$

where W_1, W_2, b_1, b_2 are the weights and biases of the feed-forward network.

SQL Grammar Masking. To enforce SQL grammar constraints, we apply a masking mechanism at each decoding step to eliminate invalid token candidates. Let $\mathcal{C}_i$ represent the set of valid tokens allowed by the SQL grammar at step i. The mask M_i is defined as:

$$M_i[t] = \begin{cases} 0, & \text{if } t \in \mathcal{C}_i, \\ -\infty, & \text{if } t \notin \mathcal{C}_i, \end{cases}$$

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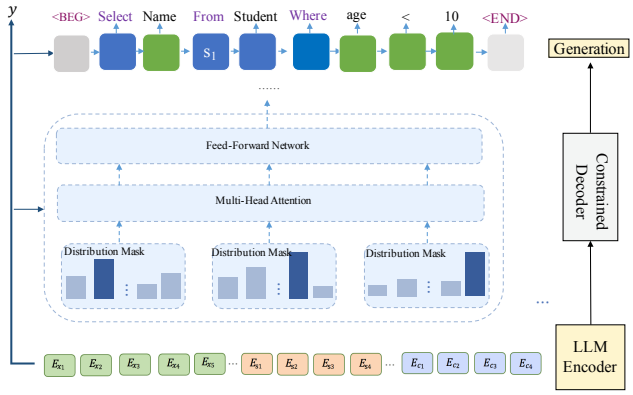

FIGURE 4. The figure illustrates the constrained decoder, which operates in three stages: the Sketch Decoder generates the high-level structure of the SQL query, the Column Prediction module identifies relevant columns from the schema, and the Detail Completion module fills in the details to produce a complete and correct SQL query. The decoder leverages embeddings from the LLM encoder, multi-head attention, and distribution masks to ensure accuracy and syntactic correctness.

where M_i is applied to the logits before the softmax operation:

$$p(y_i|y_{<i}, H_{\text{enc}}, \mathcal{C}_i) = \text{softmax}(z_i + M_i),$$

and z_i are the unnormalized logits output by the decoder.

Output Layer. The final output of the decoder is the sequence of tokens $y_{\text{sql}} = \{y_1, y_2, \ldots, y_T\}$, generated sequentially. At each step i, the token y_i is selected as:

$$y_i = \arg\max_{y \in \mathcal{C}_i} p(y|y_{<i}, H_{\text{enc}}).$$

By combining transformer layers with SQL grammar masking, the decoder ensures that the generated SQL queries are both syntactically valid and semantically accurate.

IV. EXPERIMENT

In this section, we assess the performance of our proposed model, SLENet, by comparing it against a range of baseline approaches. Additionally, we conduct a thorough ablation study and an in-depth error analysis to uncover valuable insights into the model's strengths and limitations.

A. EXPERIMENT SETUP

Dataset. We conduct our evaluation on the Spider dataset [6], a large-scale and human-annotated Text-to-SQL benchmark designed to assess models on complex SQL queries spanning 206 diverse databases. The dataset is divided into 146 training, 20 development, and 40 test databases, comprising $8,625$, $1,034$, and $2,000$ examples, respectively. Queries are classified into four levels of complexity—*easy*, *medium*, *hard*, and *extra hard*—based on the number of SQL components involved. To enhance input representation, we preprocess the natural language questions by removing stop words and applying token lemmatization. Furthermore, our model benefits from a search-based prompt mechanism that leverages external datasets, such as WikiSQL, to retrieve semantically similar examples, providing additional context to aid SQL generation. For evaluation, we adopt the SQL Exact Matching and Component Matching metrics proposed by Spider. Model predictions for the hidden test set are submitted to the official evaluation server to obtain benchmark scores, ensuring a standardized comparison with prior approaches.

Baselines. To establish the baseline, we evaluate six widely recognized models in the Text-to-SQL domain, representing a range of approaches to SQL generation. First, the sequence-to-sequence model serves as a foundational architecture for mapping natural language to structured queries, though it struggles with complex cross-domain schemas and unseen databases. Second, SQLNet [2] introduces a sketch-based approach, treating SQL generation as a slot-filling task and improving over Seq2Seq in structured scenarios. Third, IRNet [5] bridges the semantic gap between natural language and SQL by employing intermediate representations, enhancing schema comprehension. Fourth, RAT-SQL [7] incorporates relation-aware transformers to address schema complexity and improve cross-domain performance. Fifth, SmBoP [8] utilizes a semi-autoregressive decoding strategy to generate SQL queries efficiently, while ensuring syntactic correctness. Lastly, PICARD [4] introduces constrained decoding to further ensure syntactic and semantic validity in SQL generation. These baselines provide a comprehensive evaluation framework to highlight the advancements and effectiveness of SLENet.

B. EXPERIMENT RESULT

TABLE 1. Comparison of exact matching accuracy on SQL queries for SLENet and baseline models.

Approach	Dev (%)	Test (%)
Seq2Seq	1.9	3.7
SQLNet	10.9	12.4
IRNet	61.9	54.7
PICARD	69.7	65.6
SmBoP	-	71.1
SLENet	**75.2**	**74.1**

Note: SLENet achieves the highest exact matching accuracy among all approaches, significantly outperforming PICARD and SmBoP on both the development (Dev) and test sets. The best results for each category are highlighted in bold.

Comparison with Baselines. Table 1 presents the exact matching accuracy scores of SLENet and the selected baselines on the development and test sets. As shown in Table 1, SLENet achieves the highest performance among all models, with an exact matching accuracy of 75.2% on the development set and 74.1% on the test set. It surpasses SmBoP and PICARD, the next-best models, by 3.0% and 8.5% on the test set, respectively. Furthermore, SLENet significantly outperforms earlier models like Seq2Seq, SQLNet, and IRNet, demonstrating its ability to handle schema complexity and generate accurate SQL queries. These results highlight the

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TABLE 2. Effect of **LLM Enhanced Embedding (LLM-EE)** on performance improvement for different models.

Model	Without LLM-EE (%)		With LLM-EE (%)		Improvement (%)
	Dev	Test	Dev	Test	Test
Seq2Seq	1.9	3.7	40.2	40.1	+36.4
SQLNet	10.9	12.4	52.3	52.1	+39.7
IRNet	61.9	54.7	70.3	69.2	+14.5

Note: The table shows the performance on both the development (Dev) and test sets, with and without the use of LLM Enhanced Embedding (LLM-EE). Improvement is calculated based on the test set results.

TABLE 3. Impact of Transformer decoder layers on SLENet performance.

Decoder Layers	Dev (%)	Test (%)
2	71.3	70.1
4	73.5	72.4
6	**75.2**	**74.1**
8	76.3	73.2
10	76.9	71.8

Note: The results indicate that a 6-layer Transformer decoder achieves the best performance, with exact matching accuracy of 75.2% on the development set and 74.1% on the test set. Increasing the number of layers beyond 6 results in overfitting, while fewer layers lead to suboptimal performance.

robustness and effectiveness of SLENet in addressing the challenges of schema comprehension and SQL generation.

Experiment on LLM-EE. To assess the impact of LLM Enhanced Embedding (LLM-EE) on baseline models, we compared the performance of Seq2Seq, SQLNet, and IRNet with and without LLM-EE. As shown in Table 2, the inclusion of LLM-EE leads to substantial improvements in exact matching accuracy on both the development and test sets. Specifically, Seq2Seq, SQLNet, and IRNet achieve significant test set improvements of 36.4%, 39.7%, and 14.5%, respectively. These results demonstrate the effectiveness of LLM-EE in capturing contextual and structural information, enabling the models to better handle the challenges of schema complexity and diverse query semantics. The pronounced improvements in Seq2Seq and SQLNet highlight LLM-EE's ability to bridge gaps in foundational architectures, while the gains for IRNet underscore its role in enhancing already advanced models.

Experiment on Decoder Layers. To evaluate the impact of Transformer decoder layers on SLENet's performance, we vary the number of layers from 2 to 10 and analyze the results. As shown in Table 3, a 6-layer Transformer decoder achieves the best performance, with exact matching accuracy of 75.2% on the development set and 74.1% on the test set.

Interestingly, increasing the number of decoder layers beyond 6 results in a slight drop in performance, likely due to overfitting introduced by the deeper architecture. Conversely, reducing the number of layers below 6 leads to suboptimal performance, indicating that 6 layers provide the optimal trade-off between model capacity and generalization. These findings underscore the importance of careful tuning of the Transformer decoder layers to maximize SLENet's effectiveness in generating accurate and robust SQL queries.

V. RELATED WORK

NL2SQL. Text-to-SQL conversion has been a prominent area of research due to its potential to bridge natural language and structured query interfaces, making database interaction more intuitive [18] - [20]. Early approaches relied on rule-based methods and sequence-to-sequence models, which often struggled with the complexity of real-world datasets. Recent advancements, such as IRNet [5] and RAT-SQL [7], have introduced intermediate representations and relation-aware transformers to address the semantic gap between natural language and SQL syntax. These methods excel at mitigating discrepancies between the input query and database schemas. Additionally, models like SmBoP [8] and PICARD [4] introduced advanced decoding strategies, including semi-autoregressive and constrained decoding, to enhance SQL generation accuracy while ensuring syntactic correctness. Despite these innovations, challenges persist, particularly in handling diverse natural language expressions, complex multi-table queries, and unseen database schemas, as highlighted by benchmarks like Spider. Hierarchical schema representations (e.g., HSRNet) [3] and intermediate parsing techniques have been proposed to improve schema comprehension and simplify SQL generation. However, the diversity and complexity of real-world datasets continue to underscore the need for further advancements in this domain.

Large Language Models. The emergence of large language models (LLMs), such as LLaMA [13], T5 [9], Codex [10], and GPT-4 [17], has revolutionized Text-to-SQL research by leveraging pre-trained architectures and transfer learning capabilities. Among these, LLaMA stands out for its scalability and open-access nature, offering robust performance across various NLP tasks, including SQL generation, while maintaining computational efficiency. LLaMA's ability to process enriched representations of natural language queries and database schemas has enabled it to achieve competitive results on benchmarks like Spider. Recent advancements have further enhanced LLaMA's performance through the integration of external knowledge sources and prompt engineering strategies, such as search-based prompt optimization. These techniques dynamically refine input representations,

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improving the relevance and accuracy of generated SQL queries. However, like other LLMs, LLaMA faces challenges in adhering to syntactic constraints and mitigating errors without explicit structural guidance. To fully harness its potential in Text-to-SQL tasks, it is crucial to incorporate constrained decoding mechanisms and leverage external knowledge for schema comprehension and query generation.

VI. CONCLUSION

We presented SLENet, a novel approach to Text-to-SQL conversion that leverages large language models (LLMs) to enhance semantic understanding and improve SQL generation. By combining context-aware representations, a syntax-constrained SQL decoder, and search-based prompt optimization, SLENet effectively addresses challenges in schema comprehension and SQL generation. Evaluations on the Spider benchmark demonstrate that SLENet outperforms existing methods, achieving higher exact matching accuracy and handling complex SQL components. These results highlight the importance of integrating LLMs with structural constraints and external knowledge for advancing cross-domain semantic parsing. Future work will focus on enhancing robustness to real-world queries and further improving schema modeling and SQL generation accuracy.

REFERENCES

[1] V. Zhong, C. Xiong, and R. Socher, "Seq2sql: Generating structured queries from natural language using reinforcement learning," CoRR, vol. abs/1709.00103, 2017.

[2] X. Xu, C. Liu, and D. Song, "Sqlnet: Generating structured queries from natural language without reinforcement learning," arXiv preprint arXiv:1711.04436, 2017.

[3] M. Song, Z. Zhan, and H. E., "Hierarchical schema representation for text-to-sql parsing with decomposing decoding," IEEE Access, vol. 7, pp. 103706–103715, 2019, doi:10.1109/ACCESS.2019.2931464.

[4] T. Scholak, N. Schucher, and D. Bahdanau, "PICARD: Parsing incrementally for constrained auto-regressive decoding from language models," arXiv preprint arXiv:2109.05093, 2021. [Online]. Available: https://arxiv.org/abs/2109.05093.

[5] J. Guo, Z. Zhan, Y. Gao, Y. Xiao, J.-G. Lou, T. Liu, and D. Zhang, "Towards complex text-to-sql in cross-domain database with intermediate representation," arXiv preprint arXiv:1905.08205, 2019. [Online]. Available: https://arxiv.org/abs/1905.08205.

[6] T. Yu, R. Zhang, K. Yang, M. Yasunaga, D. Wang, Z. Li, J. Ma, I. Li, Q. Yao, S. Roman, Z. Zhang, and D. Radev, "Spider: A large-scale human-labeled dataset for complex and cross-domain semantic parsing and text-to-sql task," in Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing. Association for Computational Linguistics, 2018, pp. 3911–3921. [Online]. Available: http://aclweb.org/anthology/D18-1425

[7] B. Wang, R. Shin, X. Liu, O. Polozov, and M. Richardson, "RAT-SQL: Relation-aware schema encoding and linking for text-to-sql parsers," arXiv preprint arXiv:1911.04942, 2021. [Online]. Available: https://arxiv.org/abs/1911.04942.

[8] O. Rubin and J. Berant, "SmBoP: Semi-autoregressive bottom-up semantic parsing," arXiv preprint arXiv:2010.12412, 2021. [Online]. Available: https://arxiv.org/abs/2010.12412.

[9] C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu, "Exploring the limits of transfer learning with a unified text-to-text transformer," arXiv preprint arXiv:1910.10683, 2023. [Online]. Available: https://arxiv.org/abs/1910.10683.

[10] M. Chen, J. Tworek, H. Jun, Q. Yuan, H. P. de Oliveira Pinto, J. Kaplan, H. Edwards, Y. Burda, N. Joseph, G. Brockman, A. Ray, R. Puri, G. Krueger, M. Petrov, H. Khlaaf, G. Sastry, P. Mishkin, B. Chan, S. Gray, N. Ryder, M. Pavlov, A. Power, L. Kaiser, M. Bavarian, C. Winter, P. Tillet, F. P. Such, D. Cummings, M. Plappert, F. Chantzis, E. Barnes, A. Herbert-Voss, W. H. Guss, A. Nichol, A. Paino, N. Tezak, J. Tang, I. Babuschkin, S. Balaji, S. Jain, W. Saunders, C. Hesse, A. N. Carr, J. Leike, J. Achiam, V. Misra, E. Morikawa, A. Radford, M. Knight, M. Brundage, M. Murati, K. Mayer, P. Welinder, B. McGrew, D. Amodei, S. McCandlish, I. Sutskever, and W. Zaremba, "Evaluating large language models trained on code," arXiv preprint arXiv:2107.03374, 2021. [Online]. Available: https://arxiv.org/abs/2107.03374.

[11] P. Yin and G. Neubig, "A syntactic neural model for general-purpose code generation," in Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). Association for Computational Linguistics, 2017, pp. 440–450. [Online]. Available: http://aclweb.org/anthology/P17-1041

[12] P. Yin and G. Neubig, "Tranx: A transition-based neural abstract syntax parser for semantic parsing and code generation," in Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing: System Demonstrations. Association for Computational Linguistics, 2018, pp. 7–12. [Online]. Available: http://aclweb.org/anthology/D18-2002

[13] H. Touvron, T. Lavril, G. Izacard, X. Martinet, M.-A. Lachaux, T. Lacroix, B. Rozière, N. Goyal, E. Hambro, F. Azhar, A. Rodriguez, A. Joulin, E. Grave, and G. Lample, "LLaMA: Open and efficient foundation language models," arXiv preprint arXiv:2302.13971, 2023. [Online]. Available: https://arxiv.org/abs/2302.13971.

[14] W. Hwang, J. Yim, S. Park, and M. Seo, "Achieving 90% accuracy in wikisql," 2019.a

[15] A. Sherstinsky, "Fundamentals of recurrent neural network (RNN) and long short-term memory (LSTM) network," Physica D: Nonlinear Phenomena, vol. 404, p. 132306, Mar. 2020, doi:10.1016/j.physd.2019.132306.

[16] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," arXiv preprint arXiv:1706.03762, 2023. [Online]. Available: https://arxiv.org/abs/1706.03762.

[17] OpenAI, J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F. L. Aleman, D. Almeida, J. Altenschmidt, S. Altman et al., "GPT-4 technical report," arXiv preprint arXiv:2303.08774, 2024. [Online]. Available: https://arxiv.org/abs/2303.08774.

[18] T. Yu, M. Yasunaga, K. Yang, R. Zhang, D. Wang, Z. Li, and D. Radev, "Syntaxsqlnet: Syntax tree networks for complex and cross-domain text-to-sql task," in Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing. Association for Computational Linguistics, 2018, pp. 1653–1663. [Online]. Available: http://aclweb.org/anthology/D18-1193

[19] K. Xu, L. Wu, Z. Wang, M. Yu, L. Chen, and V. Sheinin, "Exploiting rich syntactic information for semantic parsing with graph-to-sequence model," in Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing. Brussels, Belgium: Association for Computational Linguistics, Oct.-Nov. 2018, pp. 918–924. [Online]. Available: https://www.aclweb.org/anthology/D18-1110

[20] I. Androutsopoulos, G. Ritchie, and P. Thanisch, "Masque/sql: An efficient and portable natural language query interface for relational databases," in Proceedings of the 6th International Conference on Industrial and Engineering Applications of Artificial Intelligence and Expert Systems, ser. IEA/AIE'93. Gordon & Breach Science Publishers, 1993, pp. 327–330. [Online]. Available: http://dl.acm.org/citation.cfm?id=1114022.1114073

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